

India Stocks Portfolio Construction

A long-only, deployable portfolio construction and backtesting framework for Indian equities. The research question is whether covariance-aware and PCA-based allocations improve the return-risk trade-off relative to equal allocation.

Markowitz Modern Portfolio Theory

Let r be the vector of asset returns, μ the expected-return vector, Σ the covariance matrix, and w the portfolio weights. Portfolio return is $R_p = w^T \mu$ and portfolio variance is $\sigma_p^2 = w^T \Sigma w$. The minimum-variance problem is to minimize $w^T \Sigma w$ subject to $w_i \geq 0$ and $\sum(w_i) = 1$. The efficient frontier represents the best achievable return for each risk level.

How the eigen portfolio connects to MPT

PCA decomposes the standardized covariance structure as $\Sigma = \sum \lambda_k v_k v_k^T$. The eigenvectors provide directions of dominant variation. Absolute loadings are normalized into long-only candidate weights, $w_{k-i} = \text{abs}(v_{k-i}) / \sum \text{abs}(v_{k-i})$. The candidate with the highest Sharpe ratio is selected, so PCA generates factor-informed portfolios while the MPT objective selects the final risk-return trade-off.

Methods

Equal weight: $w_i = 1/n$. **Minimum risk:** minimize $w^T \Sigma w$. **Optimum risk:** maximize $(R_p - R_f) / \sigma_p$ using a 1% annual risk-free rate. **Eigen:** standardize returns, compute PCA, normalize loadings, and select the best-Sharpe component portfolio.

Abstract

This project evaluates equal-weight, minimum-risk, maximum-Sharpe, and PCA eigen portfolios for Indian sector stocks. The design uses rolling one-year estimation windows and configurable monthly, quarterly, half-yearly, and yearly rebalancing. Short selling is disabled and all portfolio weights are normalized to sum to 100%.

Data and methods

Minute OHLCV records are converted into daily session-final closes. Daily returns estimate annualized return, volatility, covariance, correlation, and PCA factors. Equal weight assigns the same allocation to every stock. Minimum risk minimizes $w^T \Sigma w$. Maximum Sharpe maximizes $(R_p - R_f) / \sigma_p$. PCA decomposes standardized returns and selects the best long-only component portfolio.

Rolling backtest

At each rebalance date, only the preceding one year of data is used. The resulting weights are held until the next rebalance. Monthly, quarterly, half-yearly, and yearly schedules are supported. No future observations enter the weight calculation.

Deployment constraints

Weights are non-negative and sum to one. The framework supports maximum position weights, minimum trade sizes, turnover limits, transaction costs, slippage, cash residuals, and integer share quantities.

Evaluation

The framework reports annual return, cumulative return, CAGR, annualized volatility, Sharpe ratio, Sortino ratio, maximum drawdown, Calmar ratio, turnover, transaction costs, and final portfolio value. Equity curves are retained so drawdown metrics use daily portfolio values.

Limitations

Backtests are historical experiments, not guarantees. Risks include survivorship bias, corporate actions, stale prices, liquidity, market impact, turnover, parameter sensitivity, and omitted taxes. Production use requires adjusted prices, data validation, broker controls, and independent risk limits.

Conclusion

The project provides a reproducible long-only framework for comparing simple diversification, covariance optimization, Sharpe optimization, and PCA allocation in Indian equity sectors.